Algorithm for POS tagging

Viterbi Algorithm

- The **Viterbi algorithm** is a dynamic programming algorithm used for **Part-of-Speech (POS) tagging**, where the goal is to find the most likely sequence of POS tags for a given sentence. POS tagging is crucial in natural language processing tasks to assign grammatical labels (like noun, verb, adjective) to words in a sentence.

Key Concepts

States: These represent the possible POS tags for each word.

Observations: The words in the sentence.

Transition Probabilities: The probability of transitioning from one POS tag to another.

Emission Probabilities: The probability of a word being associated with a particular POS tag.

Start and End States: The sentence starts from an implicit "start" state, and there's also an implicit "end" state.

1. **Initialization:**
   - For each possible tag for the first word in the sentence, calculate the probability of that tag given the start state. Multiply this by the emission probability of the word given the tag.

$$viterbi[t_1, s] = P(t_1 = s|\text{start}) \times P(\text{word}_1|t_1 = s)$$

2. **Recursion:**
   - For each word in the sentence, calculate the probability for each tag by considering all possible previous tags. For each current tag `s`, select the previous tag that maximizes the product of:
     - The Viterbi score of the previous tag.
     - The transition probability from the previous tag to the current tag.
     - The emission probability of the current word for the current tag.

$$viterbi[t_i, s] = \max_{s'}\left(viterbi[t_{i-1}, s'] \times P(t_i = s|t_{i-1} = s') \times P(\text{word}_i|t_i = s)\right)$$

3. **Termination:**
    - Find the maximum probability of reaching the "end" state by considering all possible final tags.
4. **Backtracking:**
    - After filling in the Viterbi matrix, backtrack to retrieve the sequence of tags that resulted in the maximum probability.

Example (Simplified):

Consider a sentence: `"The cat sleeps"`

1. **States**: POS tags such as `NOUN`, `VERB`, `DET`.
2. **Observations**: Words: `The`, `cat`, `sleeps`.
3. **Transition Probabilities**: E.g., P(NOUN | DET), P(VERB | NOUN).
4. **Emission Probabilities**: E.g., P("cat" | NOUN), P("sleeps" | VERB).

Viterbi Table (Visualization)

Word	DET (P(DET))	NOUN (P(NOUN))	VERB (P(VERB))
The	0.5	0.1	0
cat	0	0.4	0.2
sleeps	0	0	0.8

By backtracking from the highest probability (VERB in the final state), we deduce the best tag sequence as `DET NOUN VERB`.

Corpus (Tagged Sentences):

1. "He/PRON reads/VERB books/NOUN"
2. "She/PRON eats/VERB pizza/NOUN"
3. "They/PRON enjoy/VERB football/NOUN"
4. "Books/NOUN are/VERB educational/ADJ"
5. "Pizza/NOUN is/VERB tasty/ADJ"

We will use the following **POS tags**:

- **PRON**: Pronoun (e.g., He, She, They)
- **VERB**: Verb (e.g., reads, eats, enjoy, are, is)
- **NOUN**: Noun (e.g., books, pizza, football)
- **ADJ**: Adjective (e.g., educational, tasty)

Task: POS Tagging for the sentence: **"He eats football"**.

Step 1: Compute the Probabilities

Start Probabilities ($P(t_1)$):

These are the probabilities that a particular tag is the first tag in a sentence.

$$P(\text{Tag}) = \frac{\text{Number of times Tag starts a sentence}}{\text{Total number of sentences}}$$

From the corpus:

- PRON starts 3 sentences ("He", "She", "They").
- NOUN starts 2 sentences ("Books", "Pizza").
- Total sentences = 5.

Start probabilities:

- $P(\text{PRON}) = \frac{3}{5} = 0.6$
- $P(\text{NOUN}) = \frac{2}{5} = 0.4$

Transition Probabilities ($P(t_i \mid t_{i-1})$):

These are the probabilities of transitioning from one tag to another.

$$P(\text{Tag}_2|\text{Tag}_1) = \frac{\text{Number of transitions from Tag}_1 \rightarrow \text{Tag}_2}{\text{Total transitions from Tag}_1}$$

From the corpus:

- PRON → VERB (3 times: He → reads, She → eats, They → enjoy)
- VERB → NOUN (3 times: reads → books, eats → pizza, enjoy → football)
- VERB → ADJ (2 times: are → educational, is → tasty)
- NOUN → VERB (2 times: Books → are, Pizza → is)

Total transitions from each tag:

- PRON → VERB = 3 transitions (no other transitions from PRON)
- VERB → NOUN = 3 times, VERB → ADJ = 2 times (total transitions from VERB = 5)
- NOUN → VERB = 2 transitions (no other transitions from NOUN)

Transition probabilities:

- $P(\text{VERB}|\text{PRON}) = \frac{3}{3} = 1$
- $P(\text{NOUN}|\text{VERB}) = \frac{3}{5} = 0.6$
- $P(\text{ADJ}|\text{VERB}) = \frac{2}{5} = 0.4$
- $P(\text{VERB}|\text{NOUN}) = \frac{2}{2} = 1$

Emission Probabilities (P(word | tag)):

These are the probabilities that a particular word is generated by a tag.

$$P(\text{Word}|\text{Tag}) = \frac{\text{Number of times Word is tagged as Tag}}{\text{Total number of occurrences of Tag}}$$

- **PRON**: "He" = 1, "She" = 1, "They" = 1 (Total occurrences of PRON = 3)
- **VERB**: "reads" = 1, "eats" = 1, "enjoy" = 1, "are" = 1, "is" = 1 (Total occurrences of VERB = 5
- **NOUN**: "books" = 1, "pizza" = 1, "football" = 1 (Total occurrences of NOUN = 3)
- **ADJ**: "educational" = 1, "tasty" = 1 (Total occurrences of ADJ = 2)

Emission probabilities:

- $P(\text{He}|\text{PRON}) = \frac{1}{3} = 0.33$
- $P(\text{She}|\text{PRON}) = \frac{1}{3} = 0.33$
- $P(\text{They}|\text{PRON}) = \frac{1}{3} = 0.33$
- $P(\text{reads}|\text{VERB}) = \frac{1}{5} = 0.2$
- $P(\text{eats}|\text{VERB}) = \frac{1}{5} = 0.2$
- $P(\text{football}|\text{NOUN}) = \frac{1}{3} = 0.33$
- $P(\text{tasty}|\text{ADJ}) = \frac{1}{2} = 0.5$

Step 2: Viterbi Algorithm

Given the sentence: "**He eats football**", we want to find the most likely sequence of POS tags.

Initialization (t = 1):

For the first word "He":

$$viterbi[1][\text{Tag}] = P(\text{Tag}) \times P(\text{"He"}|\text{Tag})$$

POS Tag	Start Probability	Emission Probability $P(\text{He}|\text{Tag})$	Viterbi Value
PRON	0.6	0.33	0.6 × 0.33 = 0.198
VERB	0 (cannot start)	-	-
NOUN	0.4	0 (He isn't tagged as NOUN)	0
ADJ	0 (cannot start)	-	-

Recursion (t = 2):

For the second word **"eats"**, we calculate the maximum probability for each tag by considering all possible previous tags and the corresponding transitions.

$$viterbi[2][\text{Tag}] = \max_{\text{previous state}} (viterbi[1][\text{previous state}] \times P(\text{Tag}|\text{previous state}) \times P(\text{"eats"}|\text{Tag}))$$

For VERB (possible transition from PRON):

- From PRON: $0.198 \times P(\text{VERB}|\text{PRON}) \times P(\text{"eats"}|\text{VERB}) = 0.198 \times 1 \times 0.2 = 0.0396$

| POS Tag | Previous Tag | Transition Probability | Emission Probability $P(\text{"eats"}|\text{Tag})$ | Viterbi Value | |-----------|---------------|------------------------|--|----------------| | VERB | PRON | 1.0 | 0.2 | 0.0396 |

For NOUN and ADJ:

- No valid transitions from PRON to NOUN or PRON to ADJ based on transition probabilities.

Recursion (t = 3):

For the third word "**football**", we again calculate the probabilities for each tag.

For NOUN (possible transition from VERB):

- From VERB: $0.0396 \times P(\text{NOUN}|\text{VERB}) \times P(\text{"football"}|\text{NOUN}) = 0.0396 \times 0.6 \times 0.33 = 0.00787$

For ADJ:

- No valid transitions from VERB to ADJ.

| POS Tag | Previous Tag | Transition Probability | Emission Probability $P(\text{"football"}|\text{Tag})$ | Viterbi Value | |---------|-------------|-----------------------|---|--------------| | NOUN | VERB | 0.6 | 0.33 | 0.00787 |

Step 3: Backtracking

We backtrack from the highest probability in the final step (t = 3) to determine the most probable sequence of POS tags.

Word	POS Tag
"He"	PRON
"eats"	VERB
"football"	NOUN

Final POS Tag Sequence: **PRON → VERB → NOUN**